

Computer Basics

Overview

This document provides a quick, easy-to-understand introduction to the fundamentals of computers. It's written for beginners and covers hardware, software, operating systems, storage, networking, security, and basic troubleshooting.

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1. What is a Computer?

A computer is an electronic device that processes data to perform tasks. It accepts input, processes data, stores information, and produces output. Modern computers range from tiny microcontrollers to powerful servers.

2. Hardware Components

CPU (Processor)

The Central Processing Unit executes instructions from programs. It determines how fast a computer can perform tasks. Important specs: clock speed (GHz), number of cores, and cache size.

Memory (RAM)

Random Access Memory stores data temporarily while programs run. More RAM allows more programs to run simultaneously without slowing down.

Storage (HDD / SSD)

Storage holds data permanently (until deleted).

- HDD: Mechanical, cheaper, slower
- SSD: Faster, silent, more durable

Motherboard

The main circuit board connecting all parts of the computer.

Power Supply (PSU)

Converts electricity from the wall into power the computer can use.

Input & Output Devices

Input: Keyboard, mouse, microphone, scanner

Output: Monitor, speakers, printer

GPU (Graphics Processing Unit)

Handles graphics rendering. Important for gaming, video editing, and AI work.

3. Software

System Software

Operating system and utilities that manage the computer hardware.

Application Software

Programs used by users like browsers, MS Word, Excel, media players, etc.

4. Operating Systems (OS)

An OS controls computer hardware and allows users and applications to run.

Popular OS:

- Windows
- macOS
- Linux

Mobile OS:

- Android
- iOS

5. Files and File Systems

A file is stored information.

Common file systems: NTFS, FAT32, ext4, APFS

Examples:

- .docx – Word file
- .jpg – Image
- .mp4 – Video
- .exe – Program

File paths:

- Windows: C:\Users\Name\Documents\file.txt
- Mac/Linux: /home/name/file.txt

6. Basic Computer Operations

Booting & Shutdown

- **Booting** = turning ON system
- **Shutdown** = turning OFF system safely

Installing Applications

Download from trusted source and follow installation steps.

Updates & Backups

Keep system updated and take regular backups.

7. Networking & Internet

LAN vs WAN

- LAN: Local Network (Home / Office)
- WAN: Internet (Global)

IP & DNS

- IP Address identifies a device
- DNS converts website name to IP address

Wi-Fi

Wireless internet connection through a router.

8. Security & Safety

Passwords

Use strong passwords and two-step verification (MFA).

Antivirus

Protects against viruses and malware.

Backups

Keep 2–3 copies of important files.

Safe Browsing

Avoid suspicious links & use HTTPS websites.

9. Basic Commands

Windows (CMD / PowerShell)

- dir – show files
- cd – change folder
- cls – clear screen
- ipconfig – network info

Linux/Mac Terminal

- ls – list files
- cd – change folder
- cp – copy
- mv – move
- sudo – admin command

10. Troubleshooting Checklist

1. Check power
2. Restart system
3. Check cables
4. Scan for virus
5. Reinstall driver / software
6. Restore backup

11. Glossary

Term	Meaning
BIOS	Starts the computer
Driver	Connects hardware to OS
Kernel	Heart of OS

Term	Meaning
Cloud	Online storage
RAM	Temporary memory

12. Learning Resources

- W3-Schools

Quick Tips

- ✓ Always update your computer
- ✓ Use strong passwords
- ✓ Take backups
- ✓ Learn basic commands
- ✓ Keep your system clean

Use this document as your beginner guide to Computer Basics.